

4 Resultant force along GX = $2\cos 60^\circ \times 40 = 40\text{ N}$

Hence there is an acceleration along XG since there is a resultant force of 10 N along XG. From an application of 10 N along GX, there is no net force along XG and G would move with constant speed along XG.

Answer: D

5 $E_k = \frac{1}{2}mv^2 = \frac{1}{2}p^2$ implies $p = \sqrt{2E_k m}$